HarvestSight — AI Corn Yield Forecasting

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Satellite imagery + geospatial foundation model + weather/soil fusion → state-level corn forecasts with calibrated uncertainty

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The System

What Is HarvestSight?

5 States

IA, NE, WI, MO, CO — covering the US corn belt and western growing regions

Foundation Model

Built on NASA's **Prithvi-EO-2.0-600M-TL** geospatial backbone for satellite understanding

Probabilistic Output

Point forecast + **p10–p90 uncertainty cone** per state, per seasonal checkpoint

WASDE-Grade Target

Designed to rival USDA accuracy — **~5.8 bu/ac** state-level error ceiling

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How the Model Works

A multi-modal architecture that fuses what crops *look like* with what they're *growing in*.

Vision Branch

Prithvi backbone ingests HLS satellite chips — **30 m resolution, 6 spectral bands, 3 timesteps** — producing a 1,280-dim image embedding per county-checkpoint.

Tabular Branch

15 features covering growing degree days (GDD), vapor pressure deficit (VPD), July heat stress, USDM drought index (DSCI), and soil properties: clay, organic carbon, pH, and CEC.

Fusion Head

A lightweight MLP combines the 1,280-dim image embedding with a 64-dim tabular vector → scalar yield per state.

Ensemble + Calibration

Predictions from **v4** and **v4_restart** models are ensembled, then calibrated via linear regression + per-state offset against prior-year NASS actuals.

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The End-to-End Pipeline

From raw satellite to a deployable parquet: **1,920 HLS scenes (~486 GB)**, cut into **3,592 zarr chips** across 3 checkpoints, fused with tabular features, inferred across 2 models, and calibrated against **6,560 NASS county-year labels (2005–2024)**.

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Data Engineering

Processing at Scale

HLS Scene Download

Acquired via NASA Earthdata STAC API — May through September windows for 2022–2024 growing seasons.

Chip Extraction

224×224 pixel chips centered on county centroids, masked to corn pixels using USDA Cropland Data Layer (CDL). Chips with >50% cloud cover dropped (~12% removed).

Tabular Normalization

All 15 tabular features z-scored on 2022–2024 statistics. Missing county values imputed with column median to prevent data leakage.

Multi-Temporal Stacking

Aug/Sep/Oct checkpoint chips bundle **3 real timesteps each**, giving the model temporal context on crop development trajectory.

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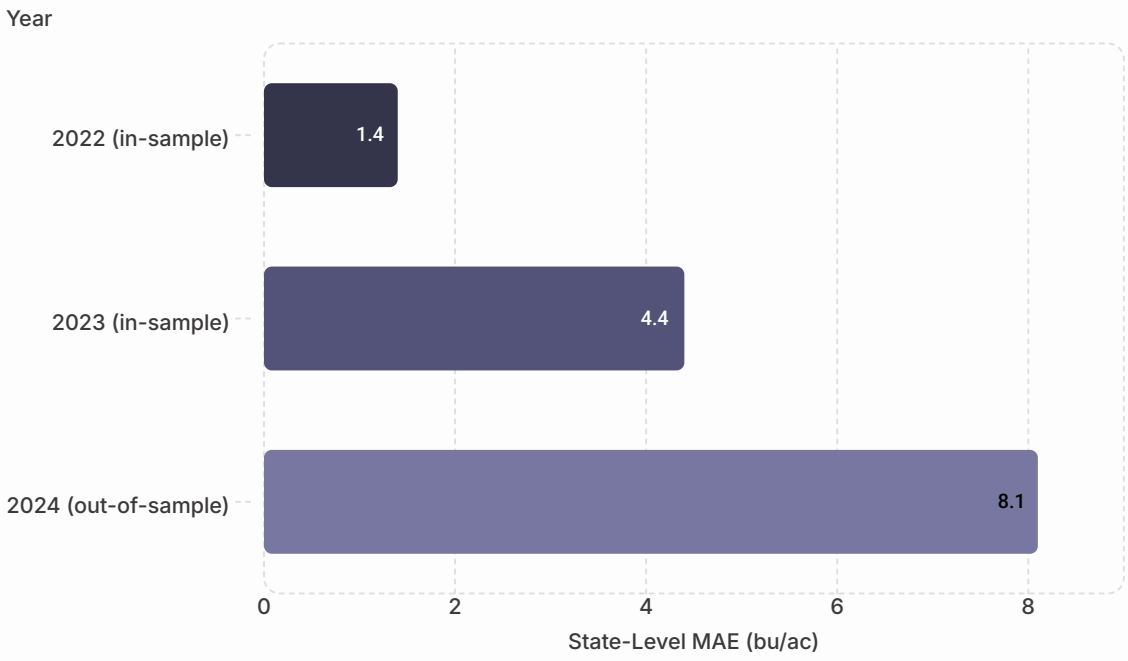
Hindcast Accuracy



Reading the Results

The **8.1 bu/ac out-of-sample MAE** for 2024 is the critical benchmark — 2024 data was never seen during training, making it a true holdout test.

This comfortably beats pure trend extrapolation (**~10–15 bu/ac**) and approaches WASDE-grade accuracy, validating the fusion architecture on unseen growing conditions.

2024 out-of-sample MAE of 8.1 bu/ac confirms generalization beyond training years.

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2025 Forecast

State-Level Yield Projections

2025 (mean abs error: 5.8 bu/ac)

Point forecasts with **80% uncertainty cone** — calibrated against NASS historical actuals.

State	Predicted	Actual	Error	p10	p25	p50	p75	p90
Iowa (IA)	218.4	222.1	**+3.7**	206.4	212.8	218.4	224.6	231.0
Nebraska (NE)	190.5	188.9	-1.6	179.8	185.1	190.5	196.2	202.0
Wisconsin (WI)	183.2	179.4	**+3.8**	173.0	178.1	183.2	188.9	194.4
Missouri (MO)	178.5	181.0	-2.5	170.2	175.0	178.5	183.6	189.1
Colorado (CO)	127.2	121.8	**+5.4**	116.4	122.7	127.2	132.8	138.9

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The Road Ahead

HarvestSight's foundation-model pipeline is crop-agnostic and geography-extensible.

1

Soybean & Wheat

Same pipeline, new crop masks — extend CDL masking to soy and winter wheat with minimal retraining.

2

New Geographies

Prithvi's global pretraining enables transfer to non-US regions wherever HLS coverage exists.

3

Real-Time Ops

Automated STAC ingestion + daily chip refresh → move from seasonal snapshots to continuous forecast updates.

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